

Capstone Exhibition Rubric

2021-2022

Poster Level of Proficiency

Criteria	Distinguished	Accomplished	Developing	Pre-Novice
Abstract (Poster) 5%	Distinguished includes all of the “Accomplished” criteria and the following ___Abstract clearly ties together the entire project from Grand Challenge to chosen solution, design requirements and prototype, testing results and conclusions ___Abstract alone generates excitement and desire to learn more about the topic ___ Writing is professional, organized and well developed.	___Abstract is a brief description of the entire work described in the poster ___ Abstract understandable without reading the entire poster ___ Includes (1) purpose of the study, (2) brief statement of what was done (without including minor details of the methods), (3) brief statement of major findings, and (4) major conclusions	___ Abstract present and relatively complete but not prepared according to all guidelines	___ No abstract
Introduction (Poster) 10%	Distinguished includes all of the “Accomplished” criteria and the following ___Moves clearly from a broad view of the Grand Challenge and research on various solutions to an increasingly narrow focus on the team’s chosen solution, design requirements, and prototype, justifying each choice. ___Provides a smooth transition from “what” choices they made and “why” they made them to the next section,	___ Connection is made to Egypt’s Grand Challenges. ___Connection is made to current semester Capstone Challenge ___ Clearly and objectively identifies the problem and summarizes prior solution attempts strengths and weaknesses. ___ Summarizes prior solution attempts strengths and weaknesses. ___Includes design requirements for a new solution that can be tested ___ Summarizes how the team’s solution was chosen and how it addresses design requirements	___ Introduction present and relatively complete but not prepared according to all guidelines	No introduction

Capstone Exhibition Rubric 2021-2022

Materials and Methods (Poster) 10%	Distinguished includes all of the “Accomplished” criteria and the following ___The methods are clear enough to permit a reader to explain the method to another professional	___Prototype materials lists and/or illustrations are summarized ___Describe how the prototype was constructed ___ A summary of Methods of the test plan for the prototype includes a summary of tests conducted and how they address design requirements	___ Materials and Method present and relatively complete but does not address all points indicated	___ No Methods or no Materials list
Results (Poster) 15%	Distinguished includes all of the “Accomplished” criteria and the following ___ Well organized results that could be handed to a new team to replicate the work with high fidelity (Supporting documentation in the Capstone Portfolio contains all data collected) ___The visual representation of the results alone (without the words in the Results section) leads the reader to a conclusion about the results.	___ All types of prototype testing results are presented, whether positive or negative ___Supporting documentation from the Capstone Portfolio is available to show data for tests or scenarios that were conducted for the prototype testing results ___ Includes a table or figure that is appropriate for the type of prototype test results being described	___ Results present and relatively complete but does not address all points indicated	___ No results
Analysis (Poster) 20%	Distinguished includes all of the “Accomplished” criteria and the following ___Analysis is clear, organized and well developed. It explains, questions or persuades. It is written to meet the needs of the intended audience.	___Discussion ties performance results (positive and negative) to the original challenge or problem being addressed and to the Grand Challenge ___Analysis ties the prototype testing results to the design requirements ___Analysis is supported by pictures, graphs, charts and other visuals, and prototype test results ___ Analysis is supported by scientific laws and theories and provides evidence of learning transfer of discipline learning outcomes	___ Analysis present and relatively complete but does not address all points indicated	___ No analysis
Conclusions 10%	___Conclusions drawn from prototype test results and analysis are also compared with the team’s research on other solutions Findings	___Conclusions are drawn from the prototype test results and analysis.	___ Conclusions and present and relatively complete but does not address all points indicated	___ No conclusions
Recommendation 5 %	___Recommendations are practical and directed towards a future research,	___ Recommendations for future study are provided, including specific ways the	___ recommendations present and relatively complete but does not	No recommendations

Capstone Exhibition Rubric 2021-2022

	engineering or policy group. Recommendations are clearly informed by the problem, their proposed solutions, and their	project could be improved in the future	address all points indicated	
Literature Cited (Poster) 5%	Distinguished includes all of the “Accomplished” criteria and the following ___At least five citations are peer-review publications.	___ Includes only sources cited in the poster text (at least 5 credible sources) Different kinds of sources (paper- books, encyclopedia - websites) not only websites ___includes only papers actually read by the students ___ Is prepared according to the American Psychological Association (APA) style guidelines	___ Literature cited present and relatively complete but does not address all points indicated	___ No literature cited (appropriate only if no citations used in the text)
(Poster) Layout Title, Name, Affiliation, Size, Graphics, Tables, Photos, Other Images (Poster) 10%	Distinguished includes all of the “Accomplished” criteria and the following ___ Writing is clear and readable ___ The poster demonstrates brevity (focused, well synthesized and straight to the point) while targeting a professional audience ___Visuals are well titled and labeled and tell clear stories with no other supporting text necessary.	___ Text is readable from a distance of about 1 meter ___Title is at top of the poster, short, descriptive of the project and easily readable at a distance of about 2 meters (words about 1.5-2.5 cm tall) ___ Includes presenter's name and school's name in a section about 20-30% smaller than the title ___Include grade level, semester and academic year (i.e., Grade 1, Semester 2, 2021/2022) ___ Illustrations, tables, figures, photographs or diagrams have unique identification numbers and a key to identify symbols ___ Text includes references to specific graphics or pictures ___ Legends include full explanation and where appropriate, color keys, scale, etc. ___ All images presented in appropriate layout and size relative to text ___ Uses forms that are common among STEM disciplines (E.g. notes, descriptive/narrative	___ Elements are present but not meeting all of the requirements in the “Accomplished” column, or some elements are missing	___ No graphics, tables, photos or other images (the poster should include at least some images)

Capstone Exhibition Rubric 2021-2022

		accounts, research reports) __Writing uses correct grammar and spelling, __Consistency between the font and background color		
Presentation 10 %	__ Uses Specific Terminology/Language __ Speaks with Appropriate Loudness, Speed, + Voice Inflection __ Uses Gestures + Body Language __ Demonstrates Enthusiasm for Subject + Interest in evaluation __ Produces Quality Visual Aids +Uses Effectively __ Provides Quality Materials + Uses Handouts Effectively	_ Introduces Self + Team (Uses Topic Transition Introductory __ Provides an Overview of Topic or Agenda of Key Ideas __ Clearly Describes +Emphasizes Key Ideas and Knowledge __ Restates objectives of Presentation __ Summarizes Key Ideas __ Reinforces/Repeats Benefits to evaluator (Learning Outcomes __ Responds to evaluator Questions +Comments	__ Elements are present but not meeting all of the requirements in the "Accomplished" column, or some elements are missing	One student monopolies presentation

Capstone Portfolio Rubric 2021-2022

Prototype Level of Proficiency				
Criteria	Distinguished	Accomplished	Developing	Pre-Novice
Construction of a testable prototype (Prototype) 50%	Distinguished includes all of the "Accomplished" criteria and the following __Prototype is directly relevant to the chosen solution (E.g. an actual water treatment step) __If software prototype (simulation) is used, modeling software such as Excel, LabView, or a programming language is demonstrated live and can be tested	__ Students can describe design requirements chosen for this prototype. Choice of design requirements is logical and well reasoned. __ A prototype has been constructed that was suitable for testing __Students can demonstrate their prototype is testable or show visual proof that prototype is testable __ If a software prototype (simulation) was used, the selection of modeling methods is logical and justified and students can identify the functions or relationships contained in	__ A prototype or model has been constructed, some justification for the selection of design requirements and modeling approach is provided, but it is incomplete	__ No prototype or model; or no evidence that constructed prototype or model would facilitate test of any of the design requirements.

Capstone Exhibition Rubric 2021-2022

		their software prototype ___The prototype materials are student purchased ___ Students design and construct the prototype— NOT done by other people ___During the testing of the prototype, all Safety Rules are followed		
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Capstone Exhibition Rubric 2021-2022

Portfolio Level of Proficiency				
Criteria	Distinguished	Accomplished	Developing	Pre-Novice
Prototype Testing (Portfolio) 20%	Distinguished includes all of the “Accomplished” criteria and the following ___Capstone Portfolio contains a test plan that clearly connects every type of test listed to a specific design requirement, and all chosen design requirements are addressed by the test plan. ___Capstone Portfolio clearly communicates how each test in the test plan isolates what is being tested and acknowledges other interfering factors that might affect the results. ___Capstone Portfolio contains measurement methods that have quantified error described explicitly (E.g. +/- 0.1 Volts) ___The Materials list is clear enough that a reader can replicate the work precisely.	___Capstone Portfolio contains a test plan for the prototype, which provides the scenarios to be tested, and how they relate to design requirements. ___ Capstone Portfolio contains a test plan that describes tests to be conducted in a thorough and clearly understandable manner ___Capstone Portfolio test plan supports repetition and testing by others. ___Capstone Portfolio contains a materials lists with cost (includes receipts if purchased) and/or illustrations, needed to replicate the prototype or model. ___ The prototype budget will not exceed 1000 LE ___Capstone Portfolio indicates that the total materials expenditures are within the budget.	___ Testing plan exists and partially describes the testing to be conducted; limited justification of why the tests were selected	___Testing plan is missing altogether, or fails to demonstrate any understanding of why tests relate to the design requirements
Data Collection and Analysis (Portfolio) 30%	Distinguished includes all of the “Accomplished” criteria and the following ___Capstone Portfolio accurately records all data collected through the project phases ___Capstone Portfolio is sufficiently well organized that it can be turned over to another group to replicate the prototype and tests. ___Capstone Portfolio lists at least 10 learning outcomes from their other subjects that they have transferred and applied in their Capstone. Each learning outcome should have one paragraph explaining how this learning outcome was transferred to their Capstone.	___ Capstone Portfolio demonstrates whether the prototype met design requirements with data from each portion of the test procedure. ___ Capstone Portfolio contains analysis supported by graphs, charts and/or other visuals. ___Capstone Portfolio lists at least 5 learning outcomes from their other subjects that they have transferred and applied in their Capstone. Each learning outcome should have one paragraph explaining how this learning outcome was transferred to their Capstone.	___Documentation is provided for some tests that were conducted, but some are not described ___ Analysis of the effectiveness of the design is generally described and has limited support using pictures, graphs, charts and other visuals	___ No documentation presented for test results, or results presented are not tied to the testing plan in any logical way